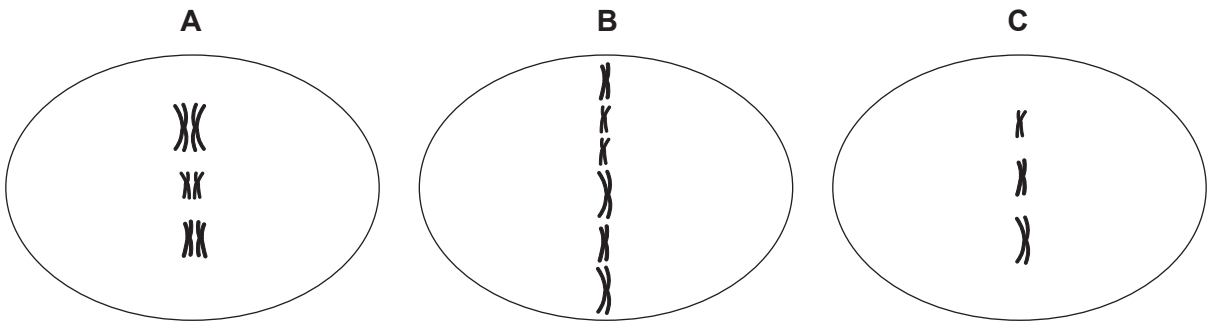


2. **Image 2.1** shows metaphase during cell division of three cells of the mosquito, *Culex pipiens*.

The mosquito has a diploid number of $2n = 6$.

Image 2.1



- (a) (i) Insert the correct letter (**A**, **B** or **C**) into the table to indicate which stage of cell division is represented by each cell. [2]

Metaphase of:	Letter
Mitosis	
Meiosis I	
Meiosis II	

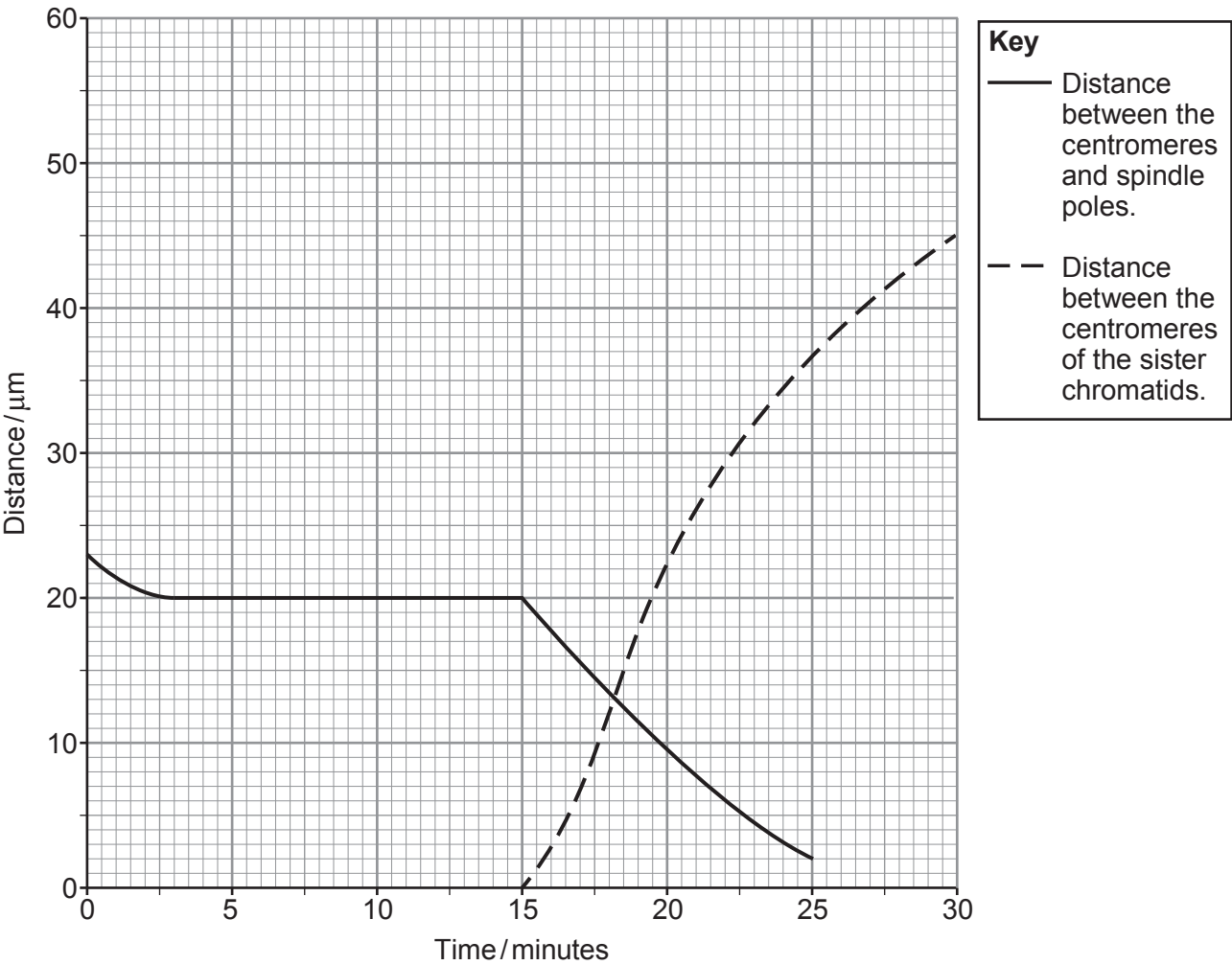
- (ii) State **two** differences in the daughter cells of *Culex pipiens* following mitosis and meiosis. [2]

Difference	Mitosis	Meiosis
1.		
2.		



(b) **Graph 2.2** shows some changes seen in a cell undergoing mitosis.

Graph 2.2



- (i) Calculate the rate of increase in distance between the centromeres of the sister chromatids between 15 and 30 minutes. [2]

Rate = $\mu\text{m minute}^{-1}$

- (ii) Suggest how **Graph 2.2** provides evidence to show what happens to the chromosomes during anaphase of mitosis. [2]

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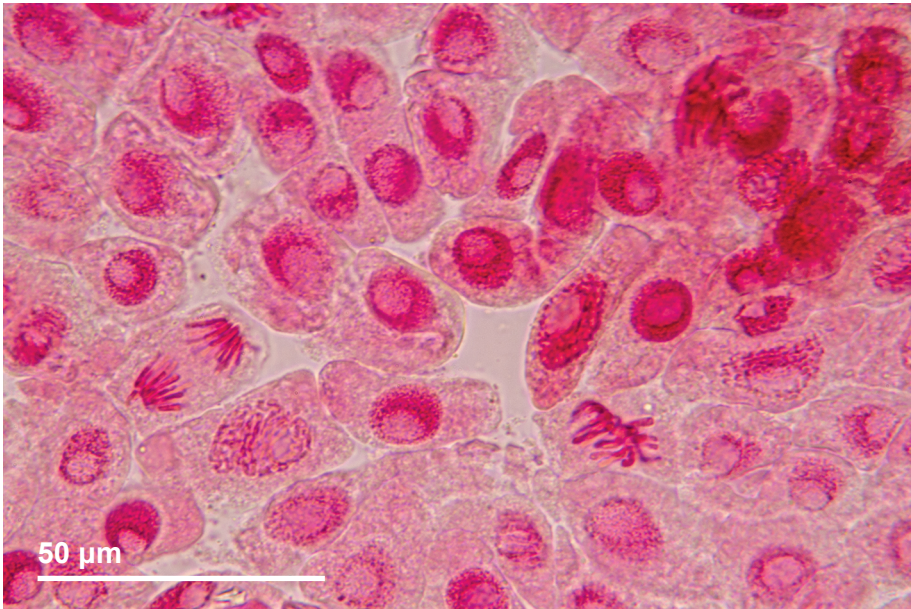
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(c) The micrograph in **Image 2.3** shows onion plant cells undergoing mitosis.

Image 2.3



(i) Use the scale bar to calculate the magnification of **Image 2.3**.

[2]

Magnification = $\times$

(ii) On **Image 2.3** label:

[1]

A one cell in anaphase

B one cell in metaphase

(iii) Suggest from which part of the onion plant the cells in **Image 2.3** were taken.
Explain your answer.

[2]

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